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Homework 2

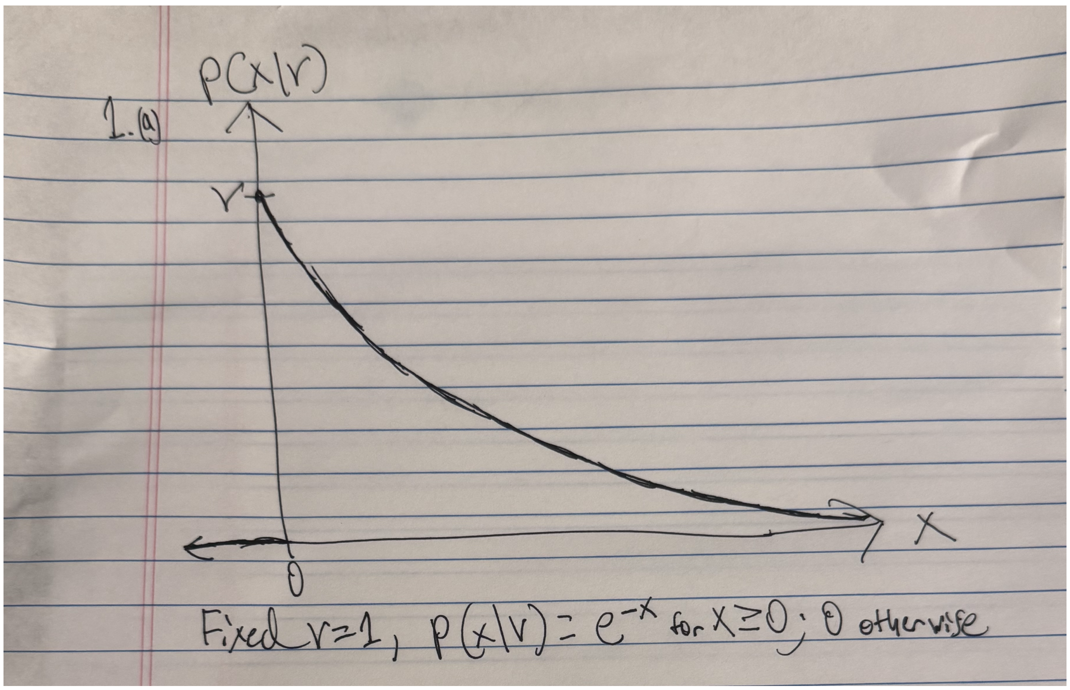
1.

Let X have an exponential distribution

$$p(x|v) = \begin{cases} ve^{-vx}, & x \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

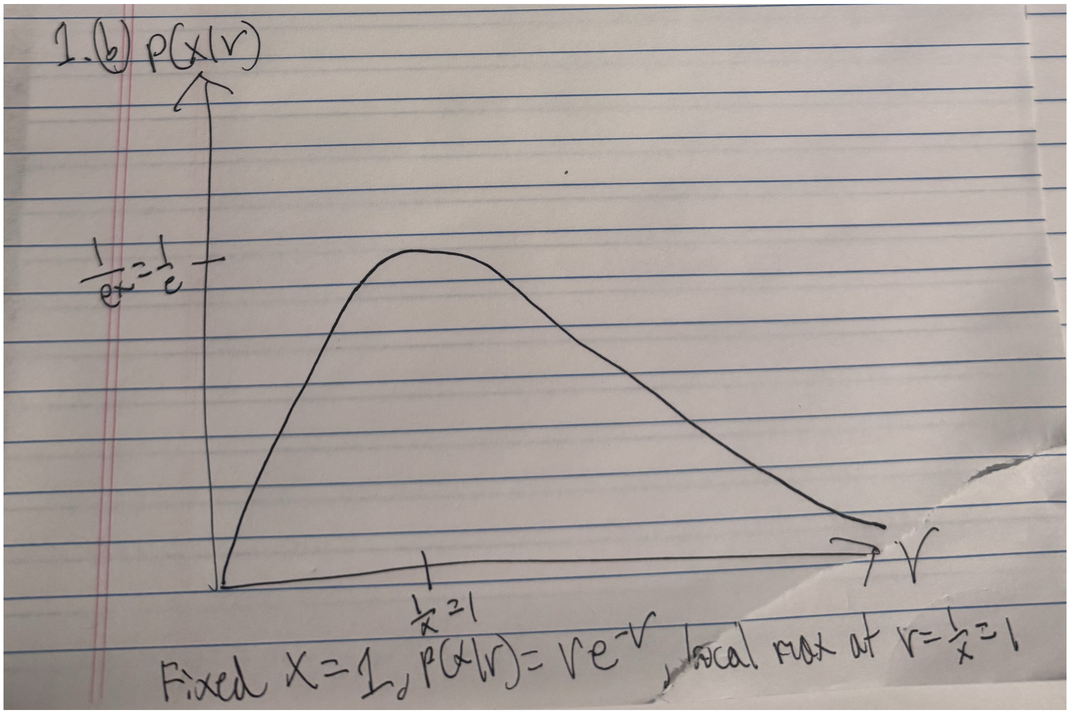
(a)

Sketch $p(x|v)$ versus x for a fixed value of the parameter v.



(b)

Sketch $p(x|v)$ versus v, $v > 0$ for a fixed value of x.



(c)

Suppose that n samples $x_1, \ldots, x_n$ are drawn independently according to $p(x|v)$. Show that the maximum likelihood estimate for v is given by

$$\hat{v} = \frac{1}{\frac{1}{n}\sum_{k=1}^{n} x_k}$$

We can derive the MLE for v directly. We first write the likelihood function. Suppose $x_1, \ldots, x_n$ are independent observations drawn from the exponential distribution, as defined by $p(x|v)$; we just multiply for a likelihood function of

$$L(v) = \prod_{k=1}^{n} ve^{-vx_k}$$

We then take the log likelihood:

$$l(v) = \ln(\prod_{k=1}^{n} ve^{-vx_k})$$

$$= \sum_{k=1}^{n} \ln(ve^{-vx_k})$$

$$= \sum_{k=1}^{n} (\ln(v) + \ln(e^{-vx_k}))$$

$$= \sum_{k=1}^{n} (\ln(v) - v x_k)$$

$$= n \ln(v) - \sum_{k=1}^{n} v x_k$$

and differentiate:

$$\frac{dl}{dv} = \frac{d}{dv}(n \ln(v) - \sum_{k=1}^{n} v x_k)$$

$$= \frac{n}{v} - \sum_{k=1}^{n} x_k$$

Verifying this is a maximum by taking the second derivative:

$$\frac{d^2 l}{dv^2} = \frac{d}{dv}(\frac{n}{v} - \sum_{k=1}^{n} x_k)$$

$$= -\frac{n}{v^2} - 0 = -\frac{n}{v^2}$$

which we see is indeed negative as both the numerator $n > 0$ and the denominator is also always positive; multiplying by this then gives us a negative value, so this is indeed a maximum. Finally, set equal to zero to solve:

$$0 = \frac{n}{v} - \sum_{k=1}^{n} x_k$$

$$\sum_{k=1}^{n} x_k = \frac{n}{v}$$

$$v = \frac{n}{\sum_{k=1}^{n} x_k}$$

So we get

$$\hat{v} = \frac{n}{\sum_{k=1}^{n} x_k} = \frac{1}{\frac{1}{n}\sum_{k=1}^{n} x_k}$$

exactly as we sought to find.

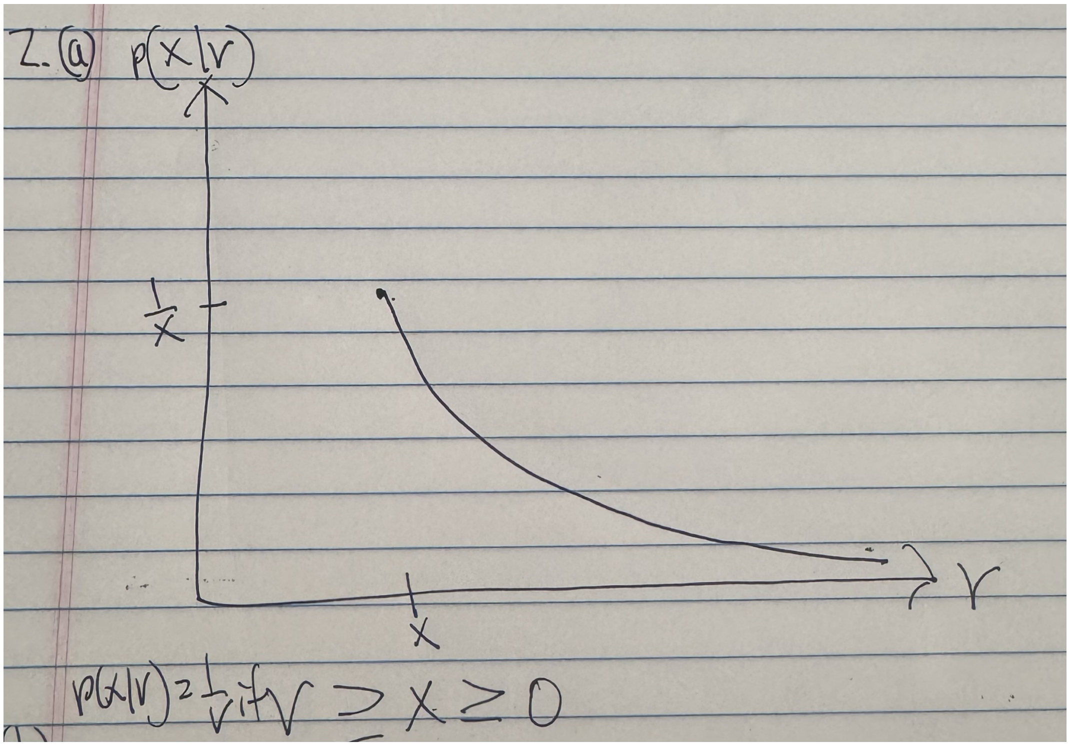
2.

Let X have a uniform distribution

$$p(x|v) = \begin{cases} \frac{1}{v}, & 0 \leq x \leq v \\ 0, & \text{otherwise.} \end{cases}$$

(a)

Sketch $p(x|v)$ versus v for an arbitrary value of x.



(b)

Suppose that n samples $x_1, \ldots, x_n$ are drawn independently according to $p(x|v)$. Show that the maximum likelihood estimate for v is $\max_k x_k$.

By definition, we have that

$$L(v) = \prod_{k=1}^{n} p(x_k|v)$$

$$L(v) = \begin{cases} \frac{1}{v^n}, & v \geq \max_k(x_k) \\ 0, & \text{otherwise.} \end{cases}$$

We note that $\frac{1}{v^n}$ continuously decreases as v increases; we therefore must choose the smallest possible value of v. We also have the restriction that v must be at least as large as the maximum x_k value, or $\max_k x_k$. We therefore have that

$$\hat{v} = \max_k x_k$$

exactly as we sought to show.

(c)

Find the method of moments estimator for v.

Let $X_1, \dots, X_n \sim \text{Uniform}(0, v)$. We therefore have the first population moment as

$$\begin{aligned} \alpha_1 = E[X] &= \int_0^v x \cdot \frac{1}{v} dx \\ &= \frac{1}{v} \cdot \frac{x^2}{2} |_0^v \\ &= \frac{1}{v} \cdot \frac{v^2}{2} - 0 \\ E[X] &= \frac{v}{2} \end{aligned}$$

We would then also have the corresponding first sample moment as

$$\hat{\alpha_1} = \frac{1}{n} \sum_{k=1}^{n} x_k = \bar{x}$$

So we finally take the method of moments estimator to be

$$\frac{\hat{v}}{2} = \bar{x}$$

$$\boxed{\hat{v} = 2\bar{x}}$$

3.

Let $\mathbf{X}$ be a binary (0, 1) vector with multivariate Bernoulli distribution

$$p(\mathbf{x}|\mathbf{v}) = \prod_{i=1}^{d} v_i^{x_i} (1 - v_i)^{1-x_i}$$

where $\mathbf{v} = (v_1, \dots, v_d)^T$ is an unknown parameter vector, v_i being the probability that $x_i = 1$. Show that the maximum likelihood estimate for $\mathbf{v}$ is

$$\hat{\mathbf{v}} = \frac{1}{n} \sum_{k=1}^{n} \mathbf{x}_k$$

We first evaluate the likelihood function, taking the product of all observations given n independent observations of $\mathbf{x}$:

$$L(\mathbf{v}) = \prod_{k=1}^{n} p(\mathbf{x}_k|\mathbf{v}) = \prod_{k=1}^{n}\prod_{i=1}^{d} v_i^{x_{ki}}(1-v_i)^{1-x_{ki}}$$

We then take the log likelihood:

$$l(\mathbf{v}) = \log(\prod_{k=1}^{n}\prod_{i=1}^{d} v_i^{x_{ki}}(1-v_i)^{1-x_{ki}})$$

$$= \sum_{k=1}^{n}\sum_{i=1}^{d} \log(v_i^{x_{ki}}(1-v_i)^{1-x_{ki}})$$

$$l(\mathbf{v}) = \sum_{k=1}^{n}\sum_{i=1}^{d}(x_{ki}\log(v_i) + (1-x_{ki})\log(1-v_i))$$

Now taking the derivative of this:

$$\frac{\partial l}{\partial v_i} = \frac{\partial}{\partial v_i}(\sum_{k=1}^{n}\sum_{i=1}^{d}(x_{ki}\log(v_i) + (1-x_{ki})\log(1-v_i)))$$

$$\frac{\partial l}{\partial v_i} = \sum_{k=1}^{n}\sum_{i=1}^{d}\frac{\partial}{\partial v_i}(x_{ki}\log(v_i) + (1-x_{ki})\log(1-v_i))$$

As we are only differentiating with respect to v_i, everything within the inner sum becomes 0 except for the v_i term itself, so this essentially simplifies to

$$\frac{\partial l}{\partial v_i} = \sum_{k=1}^{n}\frac{\partial}{\partial v_i}(x_{ki}\log(v_i) + (1-x_{ki})\log(1-v_i))$$

$$\frac{\partial l}{\partial v_i} = \sum_{k=1}^{n}(\frac{x_{ki}}{v_i} - \frac{1-x_{ki}}{1-v_i})$$

Now setting to 0 and solving:

$$0 = \sum_{k=1}^{n}\frac{x_{ki}}{v_i} - \sum_{k=1}^{n}\frac{1-x_{ki}}{1-v_i}$$

$$\sum_{k=1}^{n}\frac{x_{ki}}{v_i} = \sum_{k=1}^{n}\frac{1-x_{ki}}{1-v_i}$$

$$\frac{1}{v_i}\sum_{k=1}^{n} x_{ki} = \frac{1}{1-v_i}\sum_{k=1}^{n}(1-x_{ki})$$

$$\frac{1}{v_i}\sum_{k=1}^{n} x_{ki} = \frac{1}{1-v_i}(n-\sum_{k=1}^{n} x_{ki})$$

$$(1-v_i)\sum_{k=1}^{n} x_{ki} = v_i(n-\sum_{k=1}^{n} x_{ki})$$

$$\sum_{k=1}^{n} x_{ki} - v_i\sum_{k=1}^{n} x_{ki} = nv_i - v_i\sum_{k=1}^{n} x_{ki}$$

$$\sum_{k=1}^{n} x_{ki} = nv_i$$

$$v_i = \frac{1}{n}\sum_{k=1}^{n} x_{ki}$$

So combining this into the overall vector $\mathbf{v}$, we have that

$$\boxed{\hat{\mathbf{v}} = \frac{1}{n}\sum_{k=1}^{n} \mathbf{x}_k}$$

4.

(a)

(b)

5.

6.

(a)

(b)

(c)